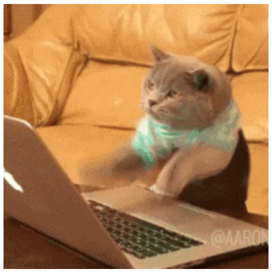

Luca Domeniconi

AI Researcher graduated in Artificial Intelligence at the University of Bologna

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Education

M.Sc. **University of Bologna**, Artificial Intelligence — Bologna, Italy, Sept 2023 – Mar 2026
- Thesis: *Using Sparse Features to Characterize and Mitigate Concept-Dependent Modality Gap in Vision-Language Models*
- Grade: 110/110 *with honors*

B.Sc. **University of Bologna**, Computer Engineering — Bologna, Italy, Sept 2020 – Oct 2023
- Grade: 109/110

Publications

How to Evaluate and Refine your CAM — Apr 2026
Luca Domeniconi, Alessandra Stramiglio, Michele Lombardi, Samuele Salti
10.1234/neurips.2023.1234 (ICPR 2026)

Experience

EPFL, Research Intern (Master's Thesis) — Lausanne, Switzerland, Oct 2025 – Mar 2026, 6 months
- Conducted research on interpretability and vision-language models for my Master's thesis project in IDIAP Lab, under the supervision of Prof. Andrea Cavallaro

Belluzzi Fioravanti High School, Computer Science Teacher — Bologna, Italy, Jan 2024 – Feb 2024, 2 months
- Taught students about computer architecture and networking using sockets

Gruppo Finmatica, Software Engineer Intern — Bologna, Italy, Mar 2023 – June 2023, 4 months
- Deployment of a web based collaborative document editor and development of a React Native prototype for barcode scanning to assist warehouse personnel in inventory management

Private Tutor — Rimini, Italy, 2019 – 2025, 6 years
- Private computer science and mathematics lessons for undergratuate and high school students

Projects

ICD Multimodal Diagnosis
Implemented a Tree-constrained multimodal ICD-9 generation pipelinefrom chest X-rays and clinical text using custom made VLMs

Semantic Segmentation of Unexpected Objects on Roads
Implemented an open-world semantic segmentation model using a pre-trained DINOv3 plus metric learning to segment known classes while detecting anomalous regions.

Multiple Couriers Optimization
Constraint programming solution to optimize the routes of multiple couriers delivering packages, written in MiniZinc and Z3.

Shelves Product Detection
Instance detection of products on shelves using only classical computer vision techniques with OpenCV

TheCatIsOnTheTablut
Agent playing Tablut (old table game) using A* search algorithm with a custom genetic heuristic, written in Python and highly optimized using Cython.

Skills

Computer Vision: Python, OpenCV, PyTorch

Machine Learning: Keras, numpy, pandas, scikit-learn, Matplotlib

Web and Mobile Development: JavaScript, NodeJS, React, React Native, Android Native

Programming Languages and Tools: Java, C, C#, \LaTeX, SQL, MATLAB, Minizinc, Unity 3D, Docker, Blender

Research Areas: Explainability, Attribution Mehthods, Vision-Language Models